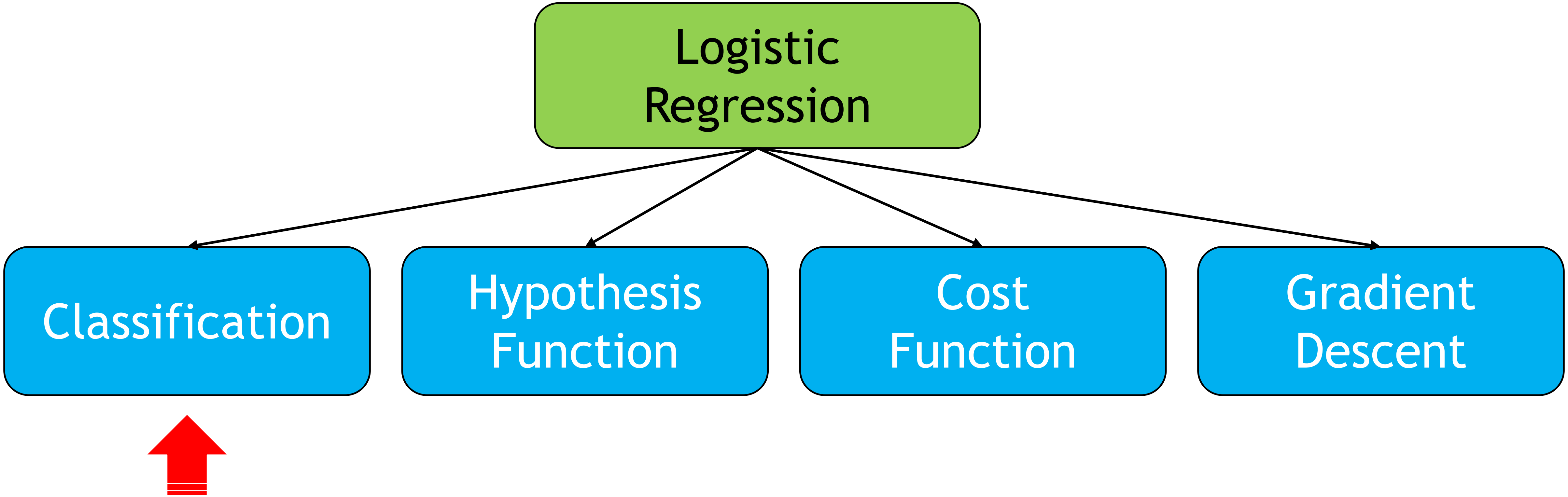
# Logistic Regression-
# Part I

Lecture 6+7:

# Outline


Logistic Regression
Classification
Hypothesis Function
Cost Function
Gradient Descent

# Example 1: Malignant or Benign

- Consider the example of recognizing whether a tumor in an input image is malignant or benign

**Input:**

**Binary Classifier**

**Output:**

Benign **0**

Malignant **1**

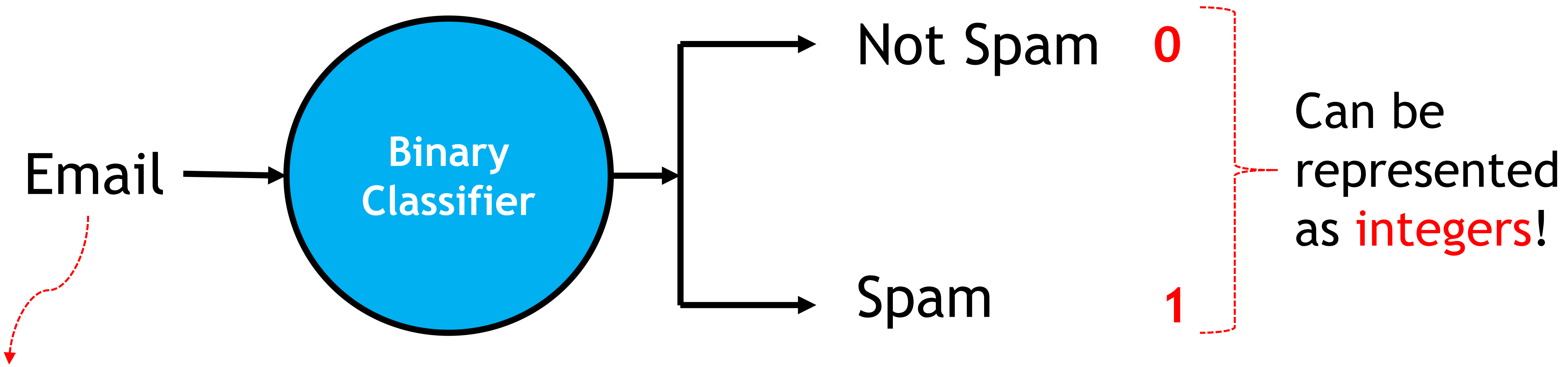
Can be represented as integers!

Can be represented as a matrix of pixels

# Example 2: Spam or Not Spam

- As another example, consider the problem of detecting whether an email is a spam or not a spam

**Input:** **Output:**




Can be represented as a vector $\mathbf{x} = [x_1, x_2, \ldots, x_d]$, with each component $x_i$ corresponding to the presence ($x_i$ = 1) or absence ($x_i$ = 0) of a particular <u>*word*</u> (or *feature*) in the email

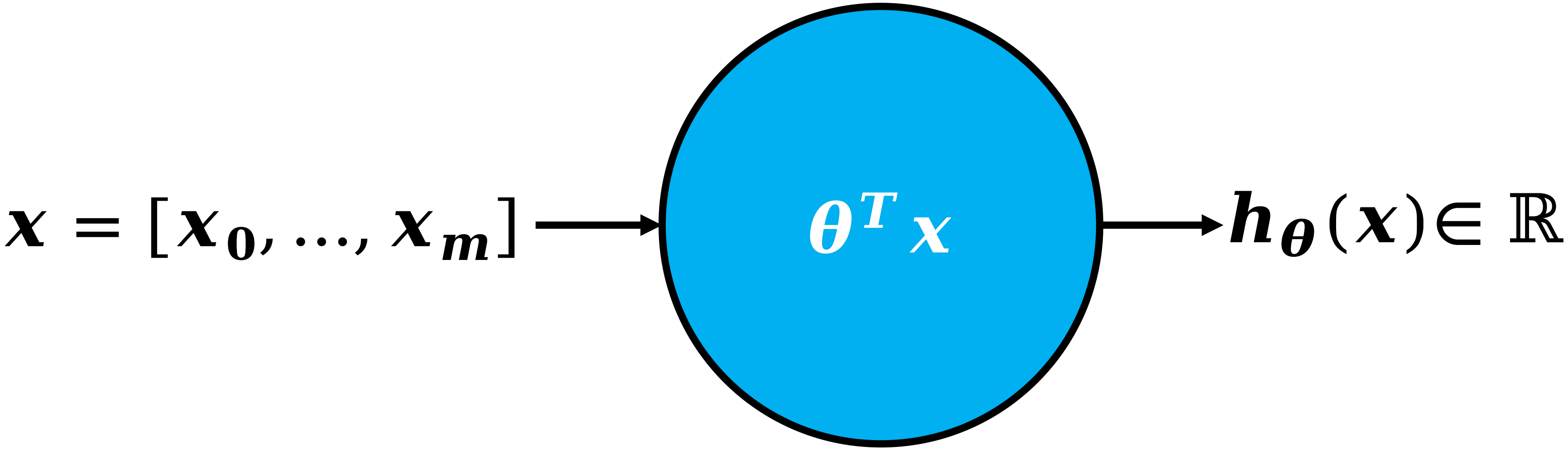
# Regression vs. Classification

- What are the possible output values of the ***linear regression*** ***model*** $\boldsymbol{h_\theta}(\boldsymbol{x}) = \boldsymbol{\theta}^{\boldsymbol{T}}\boldsymbol{x}$?

$\boldsymbol{\theta}$ is the ***parameter vector***, that is, $\boldsymbol{\theta}$ = [$\boldsymbol{\theta_0}$, $\boldsymbol{\theta_1}$, …, $\boldsymbol{\theta_m}$] (assuming $m + 1$ parameters) and $\boldsymbol{x}$ is the ***feature vector***, that is, $\boldsymbol{x}$ = [$\boldsymbol{x_0}$, $\boldsymbol{x_1}$, …, $\boldsymbol{x_m}$] (assuming $m + 1$ features and $\boldsymbol{x_0} = 1$ to account for the intercept term, namely, $\boldsymbol{\theta_0}$)

# Regression vs. Classification

- What are the possible output values of the ***linear regression model*** $\boldsymbol{h_\theta(x) = \theta^T x}$?
  - Real-valued outputs



# Regression vs. Classification

- What are the possible output values of the ***linear regression model*** $\boldsymbol{h_\theta(x) = \theta^T x}$?
  - Real-valued outputs

$$\boldsymbol{x} = [\underbrace{\mathbf{1, 0, 0, 1}}_{\text{Representing an email}}, \mathbf{0}] \rightarrow \textit{Spam Filter} \;\; \boldsymbol{\theta^T x} \rightarrow \boldsymbol{h_\theta}(\boldsymbol{x}) = \begin{bmatrix} \mathbf{3.2} \\ \mathbf{4.1} \\ \mathbf{2.9} \\ \mathbf{6.7} \\ \mathbf{1.1} \end{bmatrix} [\mathbf{1, 0, 0,}$$

Assuming $\boldsymbol{\theta} = [\mathbf{3.2, 4.1, 2.9, 6.7, 1.1}]$

# Regression vs. Classification

- What are the possible output values of the ***linear regression model*** $\boldsymbol{h_\theta}(\boldsymbol{x}) = \boldsymbol{\theta}^{\boldsymbol{T}}\boldsymbol{x}$?
    - Real-valued outputs

$\in \mathbb{R}$, which makes it a ***regression problem***

$$\boldsymbol{x} = [\mathbf{1}, \mathbf{0}, \ \mathbf{0}, \mathbf{1}, \mathbf{0}] \longrightarrow \boxed{\textit{Spam Filter} \quad \boldsymbol{\theta}^{\boldsymbol{T}}\boldsymbol{x}} \longrightarrow \boldsymbol{h_\theta}(\boldsymbol{x}) = \mathbf{3.2} + \mathbf{6.7} = \mathbf{9.}$$

**A spam or not a spam?**

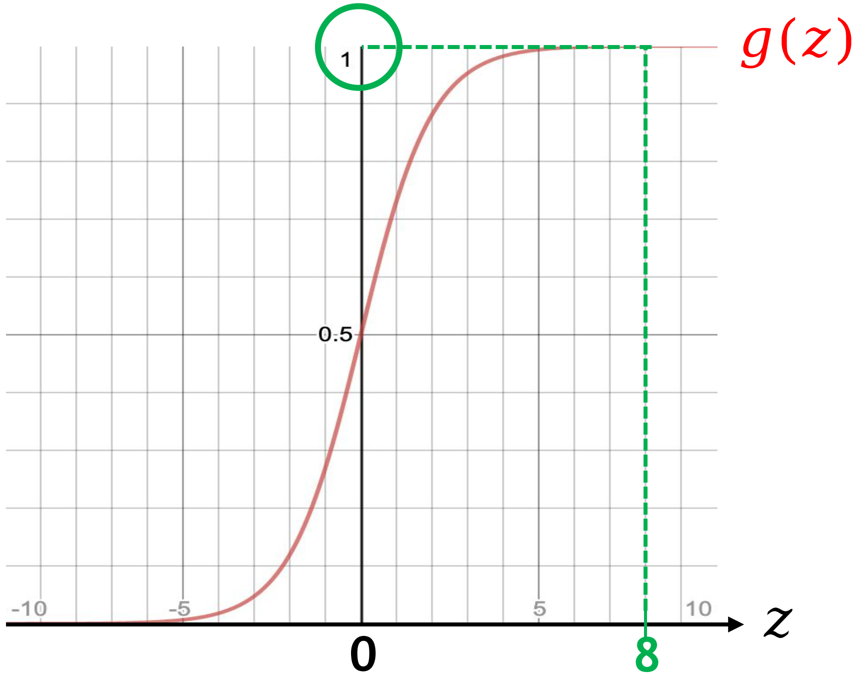
**We need a *discrete-valued* output (e.g., 1 or 0)**

# Regression vs. Classification

- How can we make the possible outputs of $\boldsymbol{h_\theta}(\boldsymbol{x}) = \boldsymbol{\theta}^{\boldsymbol{T}}\boldsymbol{x}$ discrete-valued (as opposed to real-valued)?
  - By using an ***activation function*** (e.g., ***sigmoid or logistic function***)

$$g(z) = \frac{1}{1 + e^{-z}}$$

$z \in \mathbb{R},\ but$
$g(z) \in [0,1]$




Assume a labeled example ($\boldsymbol{x}$, y)

If y = **1**, we want $g(z) \approx 1$ (i.e. we want a correct prediction)

For this to happen, $\boldsymbol{z} \gg \mathbf{0}$

# Regression vs. Classification

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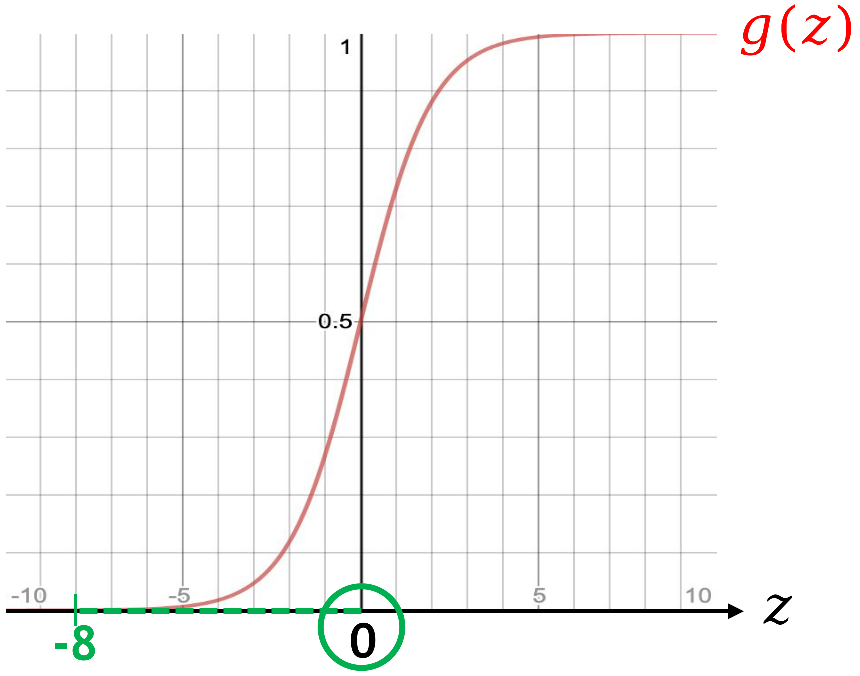


Assume a labeled example ($\boldsymbol{x}$, y)

If y = **0**, we want $g(z) \approx 0$ (i.e. we want a correct prediction)

For this to happen, $\boldsymbol{z} \ll \boldsymbol{0}$

# Regression vs. Classification

- How can we make the possible outputs of $\boldsymbol{h_\theta(x) = \theta^T x}$ discrete-valued (as opposed to real-valued)?
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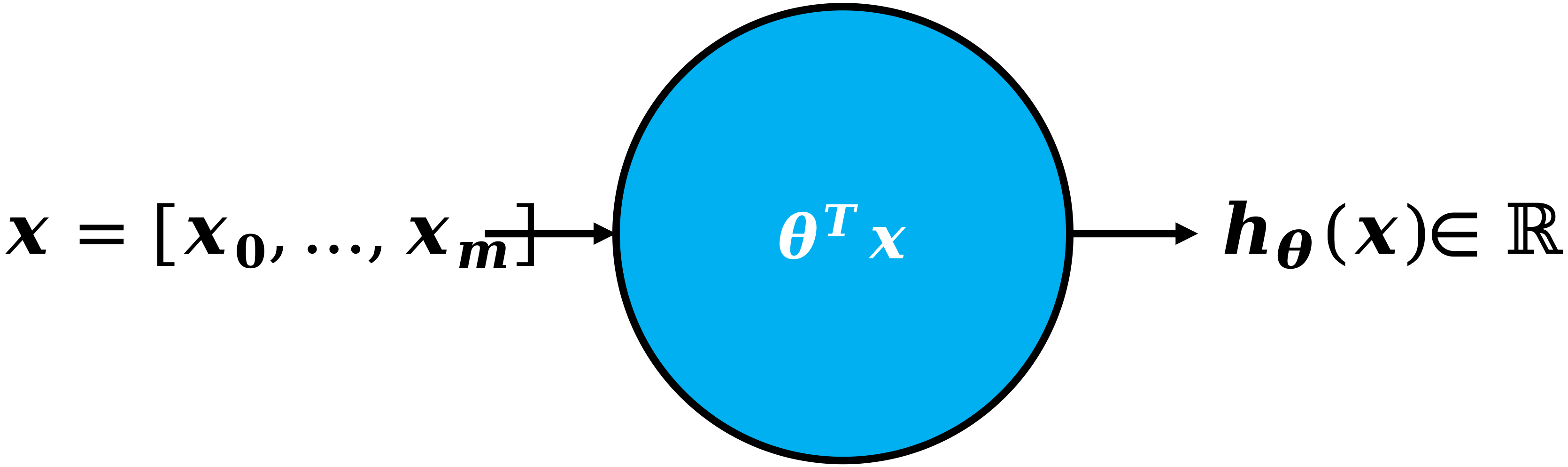


# Regression vs. Classification

- How can we make the possible outputs of $\boldsymbol{h_\theta(x) = \theta^T x}$ discrete-valued (as opposed to real-valued)?
  - By using an ***activation function*** (e.g., ***sigmoid or logistic function***)

$$\boldsymbol{x} = [\boldsymbol{x_0}, \ldots, \boldsymbol{x_m}] \rightarrow \boldsymbol{g}(\underbrace{\boldsymbol{\theta^T x}}_{\boldsymbol{z}}) \rightarrow \boldsymbol{h_\theta(x)} \in [0,1]$$

We can now apply *thresholding*-- for example, if $\boldsymbol{h_\theta(x)} < 0.5$, predict 0; otherwise, predict 1

# Regression vs. Classification

- How can we make the possible outputs of $\boldsymbol{h_\theta(x) = \theta^T x}$ discrete-valued (as opposed to real-valued)?
  - By using an ***activation function*** (e.g., ***sigmoid or logistic function***)

$$\boldsymbol{x} = [\boldsymbol{x_0}, \ldots, \boldsymbol{x_m}] \longrightarrow \boldsymbol{g}(\underbrace{\boldsymbol{\theta^T x}}_{\boldsymbol{z}}) \longrightarrow \boldsymbol{h_\theta}(\boldsymbol{x}) \in [0,1]$$

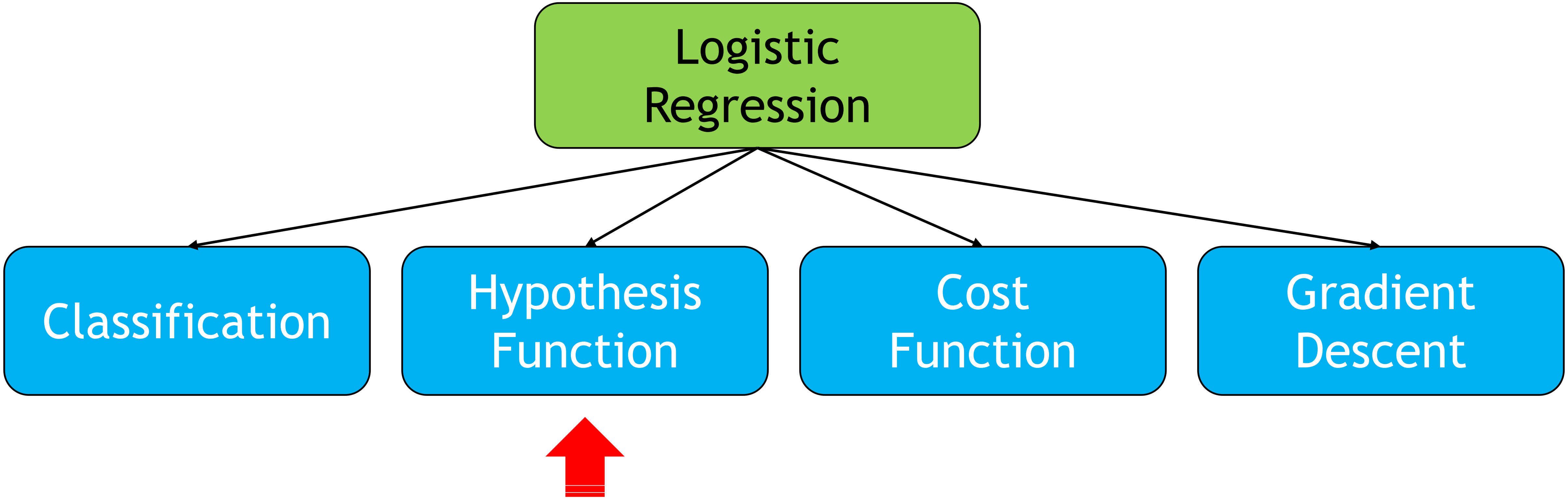
This makes it a *classification problem* (i.e., NOT a regression problem anymore) since the output renders a discrete one (*especially after applying thresholding*)

# So, What is Logistic Regression?

- Logistic regression is a machine learning algorithm that can be used to *classify* input data into discrete output (e.g., *input emails into spam or non-spam and tumour images into benign or malignant*)
  - **Note**: The word "regression" in the name does not mean that the algorithm is a regression algorithm (rather it is a classification algorithm)

- Major questions about logistic regression:
  - What is the ***hypothesis function*** (or ***model***)?
  - What is the ***cost function***?
  - How can we learn the parameters of the model?

# Outline


Logistic Regression
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# The Logistic Regression Model

- What will be the output of the model $\boldsymbol{h_\theta(x) = \theta^T x}$, where $\boldsymbol{\theta} = [\boldsymbol{\theta_0}, \ \dots, \boldsymbol{\theta_m}]$ and $\boldsymbol{x} = [\boldsymbol{x_0}, \dots, \boldsymbol{x_m}]$?
  - Real-valued

- How can we make the output of $\boldsymbol{h_\theta(x)}$ discrete?
  - By using the logistic function as follows:

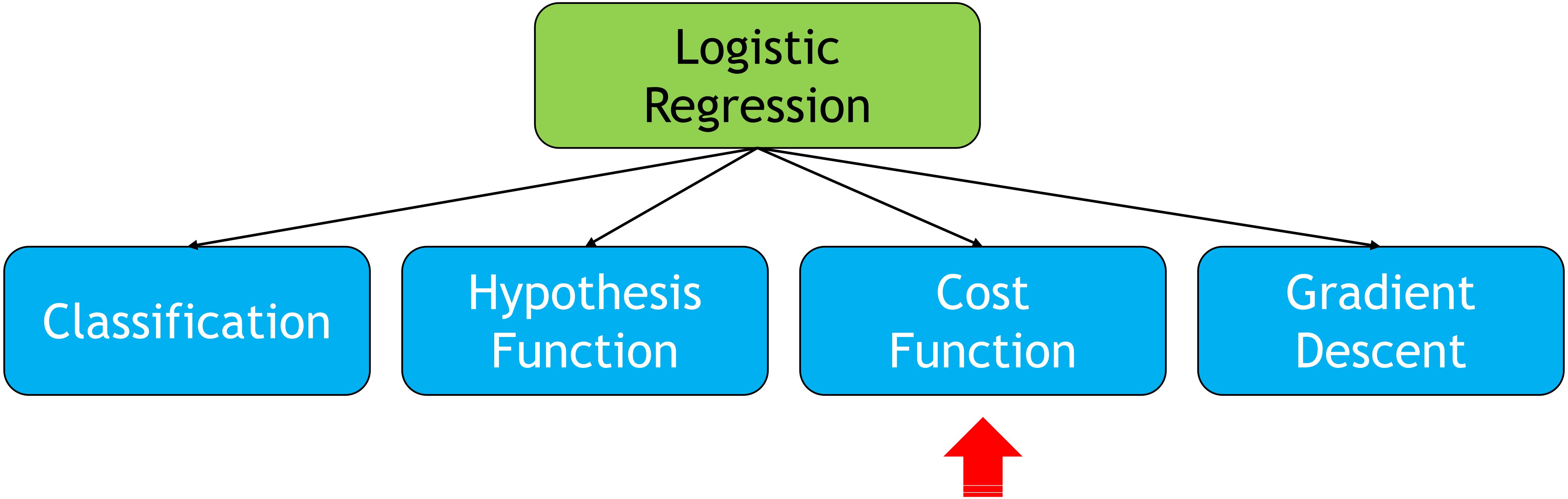
$$\boldsymbol{h_\theta(x)} = \boldsymbol{g(\theta^T x)} = \boxed{\frac{1}{1+e^{-\theta^T x}}}$$

*This is the logistic regressio model or hypothesis functio*

  - And then applying thresholding *after* learning the model to predict the output as follows:

$$\begin{cases} \boldsymbol{if\ h_\theta(x) < 0.5\ \mathbf{predict}\ 0} \\ \boldsymbol{if\ h_\theta(x) \geq 0.5\ \mathbf{predict}\ 1} \end{cases}$$

# Outline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# Towards Identifying the Logistic Regression Cost Function

- How to learn a *logistic regression model* $\boldsymbol{h_\theta}(\boldsymbol{x}) = \boldsymbol{g}(\boldsymbol{\theta}^T\boldsymbol{x})$, where $\boldsymbol{\theta} = [\boldsymbol{\theta_0}, \ldots, \boldsymbol{\theta_m}]$ and $\boldsymbol{x} = [\boldsymbol{x_0}, \ldots, \boldsymbol{x_m}]$?
  - Perhaps, by minimizing ***Mean Squared Error (MSE)***. That is:

$$\text{minimize} \frac{1}{2n} \sum_{i=1}^{n} \left(y'^{(i)} - y^{(i)}\right)^2$$

$$\equiv$$

$$\underset{\theta}{\text{minimize}} \boxed{\frac{1}{2n} \sum_{i=1}^{n} \left((g(\theta^T x))^{(i)} - y^{(i)}\right)^2}$$

***Cost function*** $\boldsymbol{J(\theta)}$

$$\equiv$$

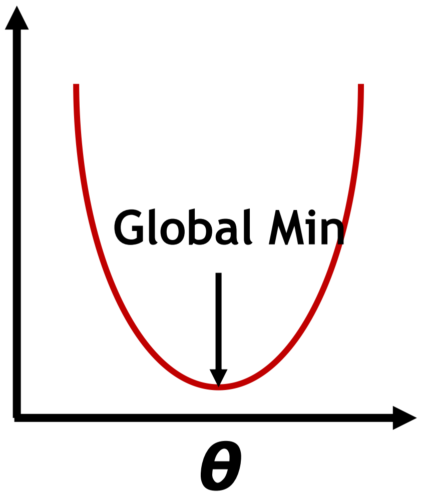
$$\underset{\theta}{\text{minimize}}\ \boldsymbol{J(\theta)}$$

# Towards Identifying the Logistic Regression Cost Function

- How to learn a *logistic regression model* $\boldsymbol{h_\theta(x)} = \boldsymbol{g(\theta^T x)}$, where $\boldsymbol{\theta} = [\boldsymbol{\theta_0}, \ldots, \boldsymbol{\theta_m}]$ and $\boldsymbol{x} = [\boldsymbol{x_0}, \ldots, \boldsymbol{x_m}]$?
  - Perhaps, by minimizing *Mean Squared Error (MSE)*. That is:

**Convex**

**Easy for Gradient Descent to locate!**

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$$\equiv$$

$$\underset{\theta}{\text{minimize}}\, \boldsymbol{J(\theta)}$$

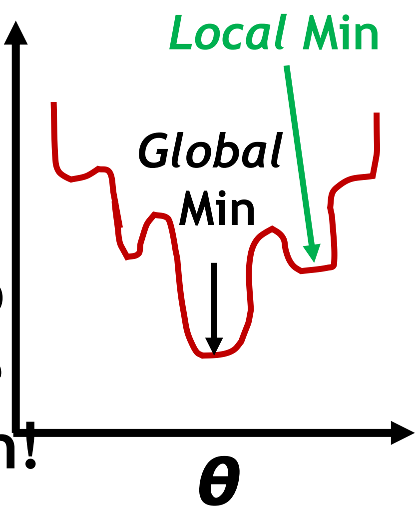
Unfortunately, if we plot this cost function, it will turn out to be “**non-convex**

# Towards Identifying the Logistic Regression Cost Function

- How to learn a *logistic regression model* $\boldsymbol{h_\theta}(\boldsymbol{x}) = \boldsymbol{g}(\boldsymbol{\theta}^{\boldsymbol{T}}\boldsymbol{x})$, where $\boldsymbol{\theta} = [\boldsymbol{\theta_0}, \ldots, \boldsymbol{\theta_m}]$ and $\boldsymbol{x} = [\boldsymbol{x_0}, \ldots, \boldsymbol{x_m}]$?
  - Perhaps, by minimizing ***Mean Squared Error (MSE)***. That is:

**Gradient Descent might get stuck at a *local* min and fail to locate the *global* min!**

**Non-Convex**




$$\text{minimize} \frac{1}{2n} \sum_{i=1}^{n} \left(y'^{(i)} - y^{(i)}\right)^2$$

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$$\underset{\theta}{\text{minimize}}\ \boldsymbol{J}(\boldsymbol{\theta})$$

Unfortunately, if we plot this cost function, it will turn out to be "**non-convex**

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- How to learn a *logistic regression model* $\boldsymbol{h_\theta}(\boldsymbol{x}) = \boldsymbol{g}(\boldsymbol{\theta}^{\boldsymbol{T}}\boldsymbol{x})$, where $\boldsymbol{\theta} = [\boldsymbol{\theta_0}, \ \ldots, \boldsymbol{\theta_m}]$ and $\boldsymbol{x} = [\boldsymbol{x_0}, \ldots, \boldsymbol{x_m}]$?
  - Perhaps, by minimizing ***Mean Squared Error (MSE)***. That is:

We need a cost function that is *convex*, hence, being more suitable for Gradient Descent

$$\text{minimize} \frac{1}{2n} \sum_{i=1}^{n} \left(y'^{(i)} - y^{(i)}\right)^2$$

$$\equiv$$

$$\underset{\theta}{\text{minimize}} \frac{1}{2n} \sum_{i=1}^{n} \left((g(\theta^T x))^{(i)} - y^{(i)}\right)^2$$

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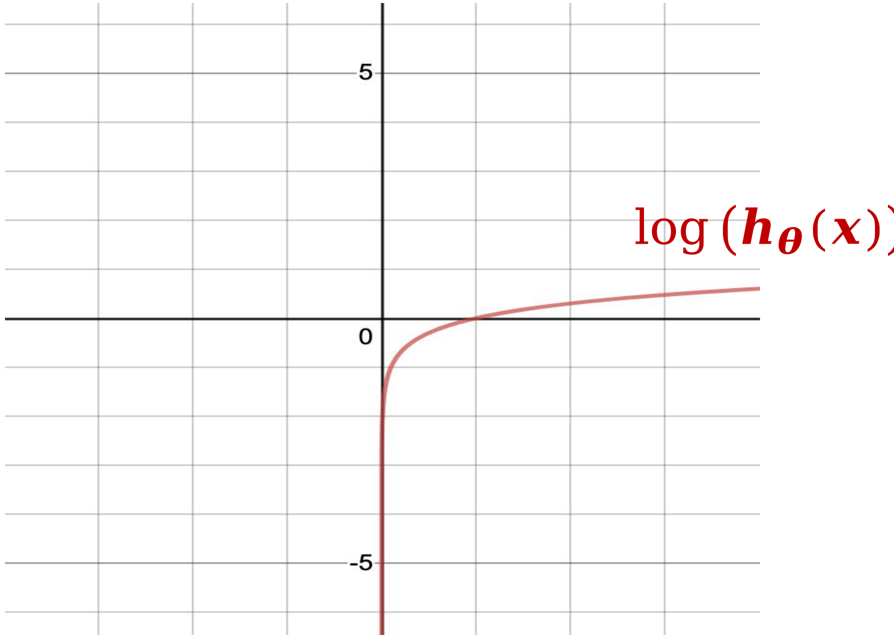
Unfortunately, if we plot this cost function, it will turn out to be "**non-convex**

# The Logistic Regression Cost Function

- How to learn a *logistic regression model* $\boldsymbol{h_\theta(x)} = \boldsymbol{g(\theta^T x)}$, where $\boldsymbol{\theta} = [\boldsymbol{\theta_0}, \ldots, \boldsymbol{\theta_m}]$ and $\boldsymbol{x} = [\boldsymbol{x_0}, \ldots, \boldsymbol{x_m}]$?
  - Let us try a different cost function. That is:

$$\text{Cost}(\boldsymbol{h_\theta(x)}, \text{y}) = \begin{cases} -\log(\boldsymbol{h_\theta(x)}) \ \boldsymbol{if} \ \boldsymbol{y = 1} \\ -\log(\boldsymbol{1 - h_\theta(x)}) \ \boldsymbol{if} \ \boldsymbol{y = 0} \end{cases}$$



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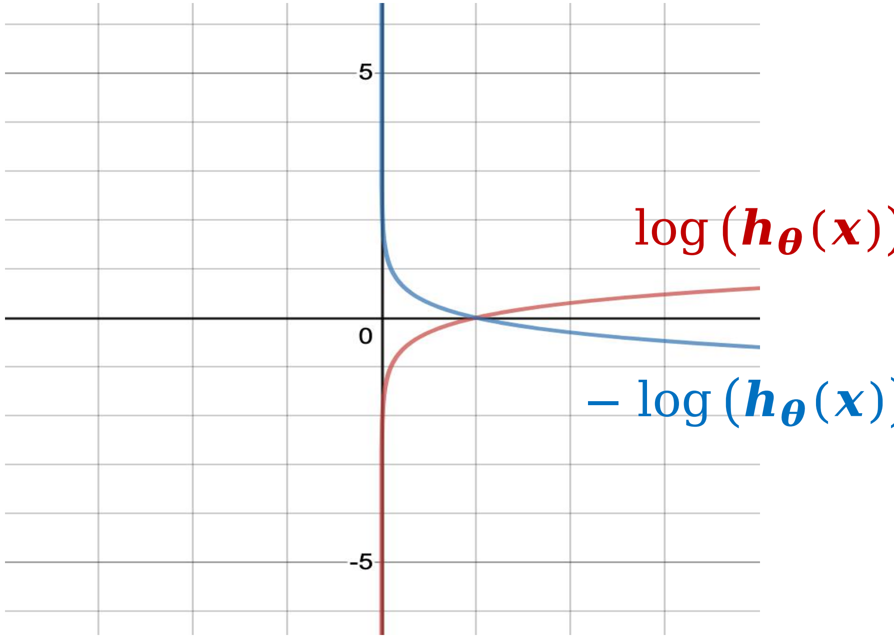

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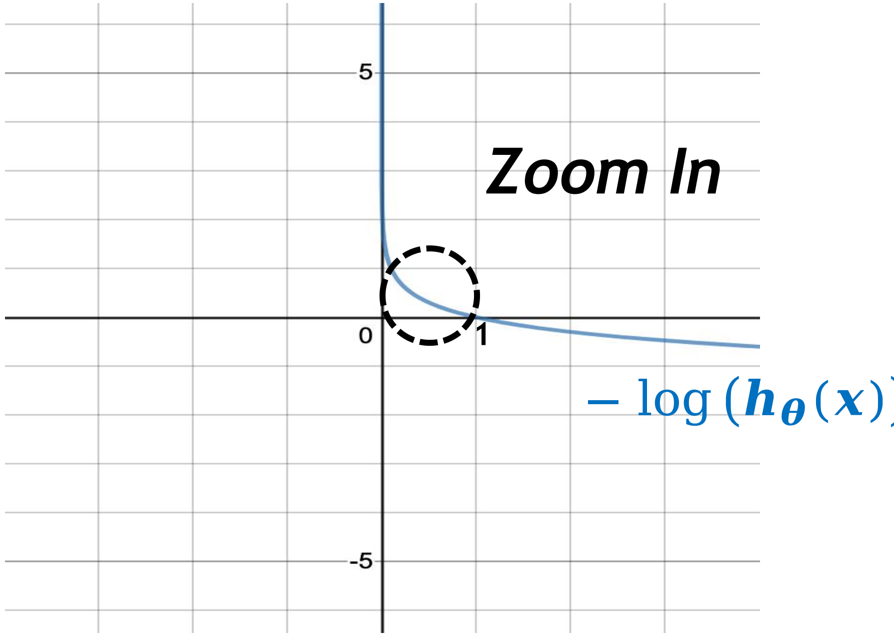

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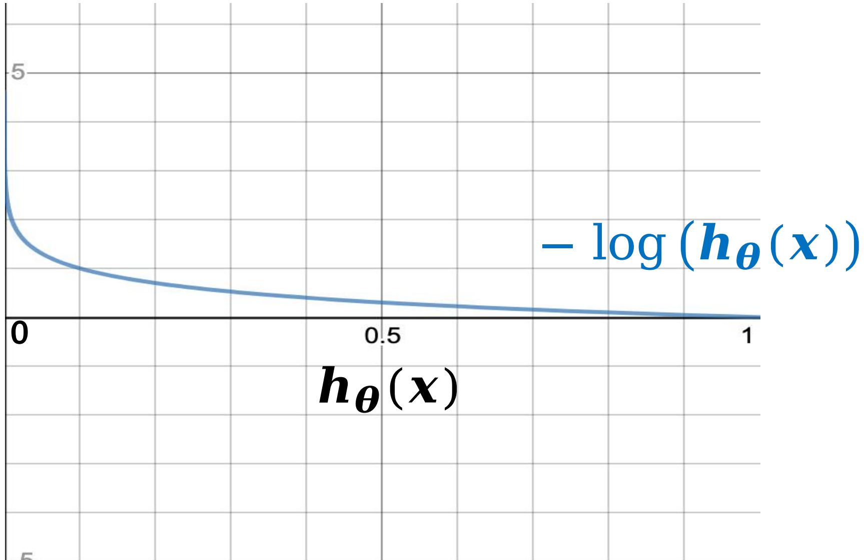


$$\boldsymbol{If}\ \ \boldsymbol{y} = \boldsymbol{1} \begin{cases} \text{As } \boldsymbol{h_\theta}(\boldsymbol{x}) \quad 0,\ -\log(\boldsymbol{h_\theta}(\boldsymbol{x})) \quad \infty \\ \text{As } \boldsymbol{h_\theta}(\boldsymbol{x}) \quad 1,\ -\log(\boldsymbol{h_\theta}(\boldsymbol{x})) \quad 0 \text{ (i.e., cost} \quad 0) \end{cases}$$

# The Logistic Regression Cost Function

- How to learn a *logistic regression model* $\boldsymbol{h_\theta(x)} = \boldsymbol{g(\theta^T x)}$, where $\boldsymbol{\theta} = [\boldsymbol{\theta_0}, \ldots, \boldsymbol{\theta_m}]$ and $\boldsymbol{x} = [\boldsymbol{x_0}, \ldots, \boldsymbol{x_m}]$?
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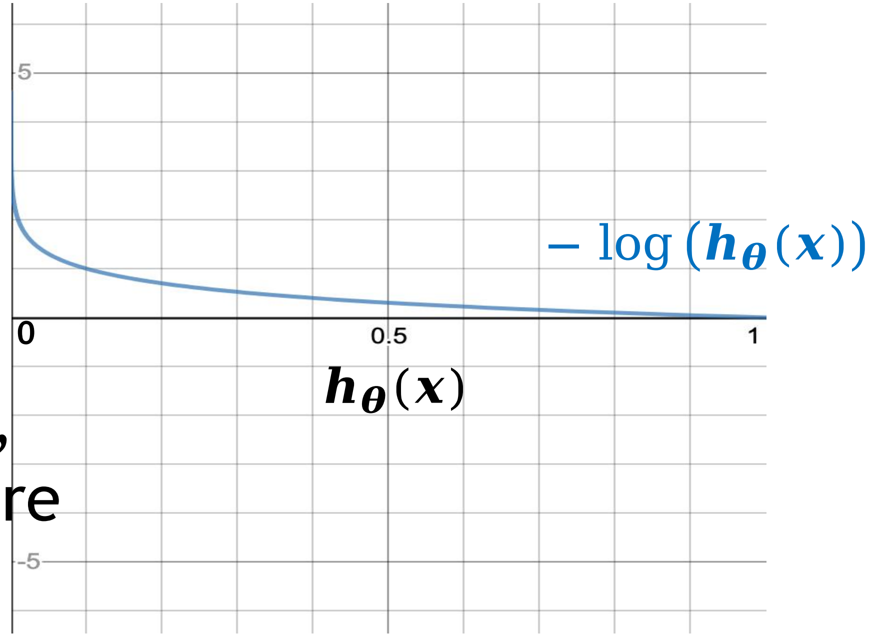


This captures the intuition that if $\boldsymbol{h_\theta(x) = 0}$ (i.e., what we will predict is 0), but $\boldsymbol{y = 1}$ (hence, we are mispredicting!), we shall penalize the learning algorithm by a very large cost!

# The Logistic Regression Cost Function

- How to learn a *logistic regression model* $\boldsymbol{h_\theta}(\boldsymbol{x}) = \boldsymbol{g}(\boldsymbol{\theta}^T\boldsymbol{x})$, where $\boldsymbol{\theta} = [\boldsymbol{\theta_0}, \ldots, \boldsymbol{\theta_m}]$ and $\boldsymbol{x} = [\boldsymbol{x_0}, \ldots, \boldsymbol{x_m}]$?
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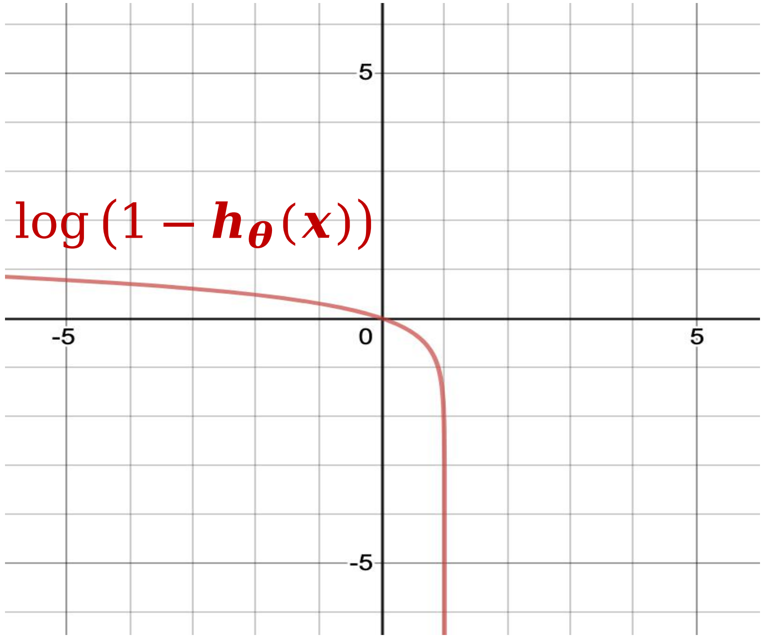

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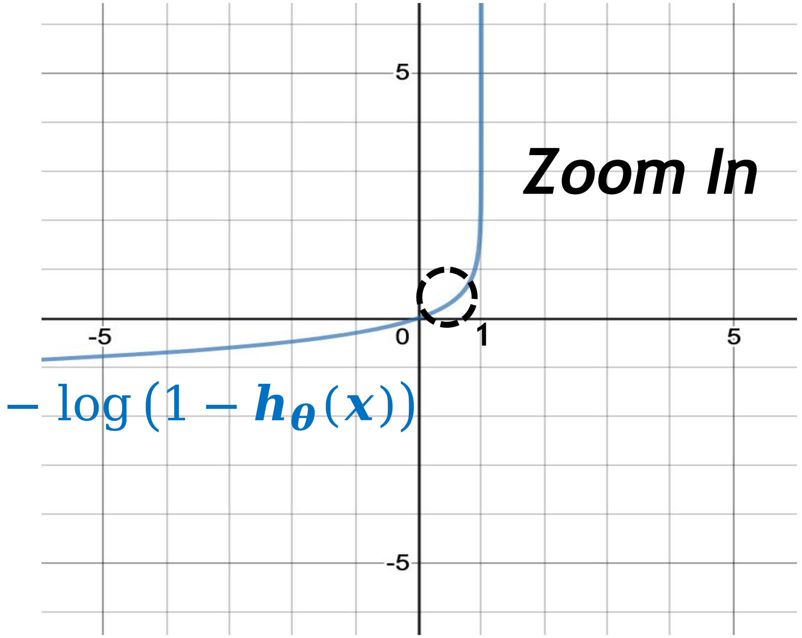

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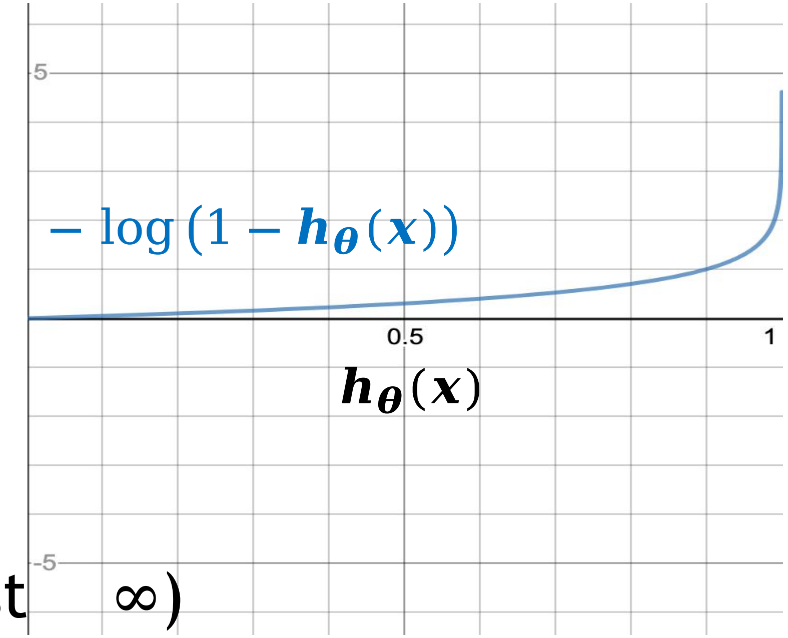


$$\boldsymbol{If \;\; y = 0} \begin{cases} \text{As } \boldsymbol{h_\theta(x)} \quad 0, \; -\log(\boldsymbol{h_\theta(x)}) \quad 0 \\ \text{As } \boldsymbol{h_\theta(x)} \quad 1, \; -\log(\boldsymbol{h_\theta(x)}) \quad \infty \text{ (i.e., cost } \quad \infty) \end{cases}$$

# The Logistic Regression Cost Function

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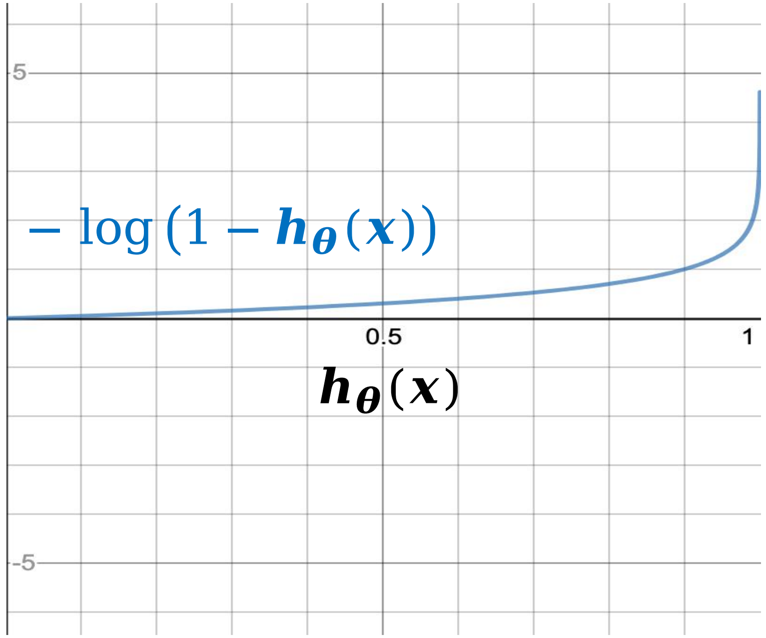


This captures the intuition that if $\boldsymbol{h_\theta(x) = 1}$ (i.e., what we will predict is 1), but $\boldsymbol{y = 0}$ (i.e., we are mispredicting!), we shall penalize the learning algorithm by a very large cost!

# The Logistic Regression Cost Function

- How to learn a *logistic regression model* $\boldsymbol{h_\theta(x)} = \boldsymbol{g(\theta^T x)}$, where $\boldsymbol{\theta} = [\boldsymbol{\theta_0}, \ \ldots, \boldsymbol{\theta_m}]$ and $\boldsymbol{x} = [\boldsymbol{x_0}, \ldots, \boldsymbol{x_m}]$?
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*Equivalent To*

$$\text{Cost}(\boldsymbol{h_\theta(x)}, \ \mathrm{y}) = y\log(\boldsymbol{h_\theta(x)}) - (1 - y)\log(\boldsymbol{1} - \ \boldsymbol{h_\theta(x)})$$

# The Logistic Regression Cost Function

- How to learn a *logistic regression model* $\boldsymbol{h_\theta}(\boldsymbol{x}) = \boldsymbol{g}(\boldsymbol{\theta}^T\boldsymbol{x})$, where $\boldsymbol{\theta} = [\boldsymbol{\theta_0}, \ldots, \boldsymbol{\theta_m}]$ and $\boldsymbol{x} = [\boldsymbol{x_0}, \ldots, \boldsymbol{x_m}]$?
  - Let us try a different cost function. That is:

$$\text{Cost}(\boldsymbol{h_\theta}(\boldsymbol{x}), \text{y}) = y\log(\boldsymbol{h_\theta}(\boldsymbol{x})) - (1-y)\log(\boldsymbol{1} - \boldsymbol{h_\theta}(\boldsymbol{x}))$$

This function still assumes real-valued outputs for $\boldsymbol{h_\theta}(\boldsymbol{x})$ (i.e., still entails *a regression problem*), while logistic regression should predict discrete values (i.e., logistic regression is a *classification problem*)

# The Logistic Regression Cost Function

- How to learn a *logistic regression model* $\boldsymbol{h_\theta}(\boldsymbol{x}) = \boldsymbol{g}(\boldsymbol{\theta}^T\boldsymbol{x})$, where $\boldsymbol{\theta} = [\boldsymbol{\theta_0}, \ldots, \boldsymbol{\theta_m}]$ and $\boldsymbol{x} = [\boldsymbol{x_0}, \ldots, \boldsymbol{x_m}]$?
  - Let us try a different cost function. That is:

$$\underbrace{\mathrm{Cost}(\boldsymbol{h_\theta}(\boldsymbol{x}), \mathrm{y})}\, = y\log\left(\boldsymbol{h_\theta}(\boldsymbol{x})\right) - (1-y)\log\left(\boldsymbol{1} - \boldsymbol{h_\theta}(\boldsymbol{x})\right)$$

We still need to apply to it the logistic function:

$$\boldsymbol{g}(z) = \frac{1}{1+e^{-z}}$$

# The Logistic Regression Cost Function

- How to learn a *logistic regression model* $\boldsymbol{h_\theta}(\boldsymbol{x}) = \boldsymbol{g}(\boldsymbol{\theta}^T\boldsymbol{x})$, where $\boldsymbol{\theta} = [\boldsymbol{\theta_0}, \ \ldots, \boldsymbol{\theta_m}]$ and $\boldsymbol{x} = [\boldsymbol{x_0}, \ldots, \boldsymbol{x_m}]$?
  - By minimizing the following cost function:

$$\text{Cost}(\boldsymbol{h_\theta}(\boldsymbol{x}), \ y) = -y\log\left(\boldsymbol{h_\theta}(\boldsymbol{x})\right) - (1-y)\log\left(\mathbf{1} - \ \boldsymbol{h_\theta}(\boldsymbol{x})\right)$$

$$= -y\log\left(\boldsymbol{g}(\boldsymbol{\theta}^T\boldsymbol{x})\right) - (1-y)\log\left(1 - \boldsymbol{g}(\boldsymbol{\theta}^T\boldsymbol{x})\right)$$

$$= -y\log\left(\frac{1}{1+e^{-\boldsymbol{\theta}^T x}}\right) - (1-y)\log\left(1 - \frac{1}{1+e^{-\boldsymbol{\theta}^T x}}\right)$$

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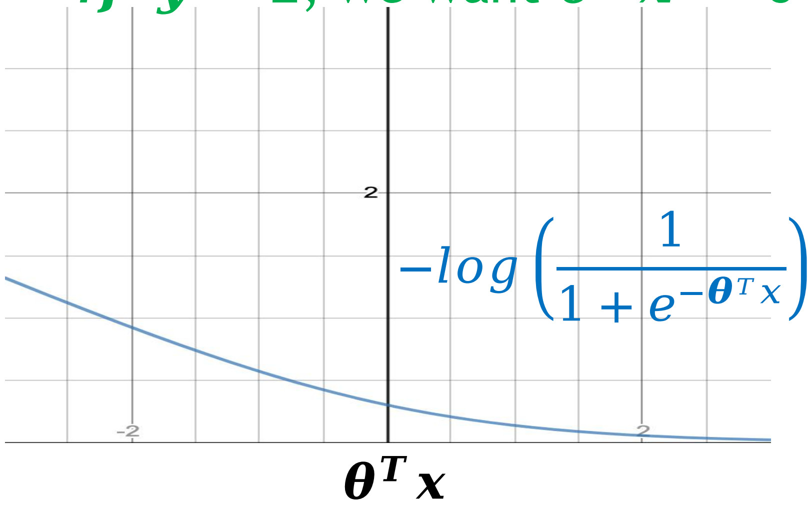
Cost $(\boldsymbol{h_\theta}(\boldsymbol{x}), \mathrm{y})$ = $-y\log\left(\frac{1}{1+e^{-\boldsymbol{\theta}^T x}}\right) - (1-y)\log\left(1 - \frac{1}{1+e^{-\boldsymbol{\theta}^T x}}\right)$

*If* $y = \mathbf{1}$, we want $\boldsymbol{\theta}^T\boldsymbol{x}$ >> 0




*If* $y = \mathbf{0}$, we want $\boldsymbol{\theta}^T\boldsymbol{x}$ << 0